

Dr Jade Quinton Maggs

Fisheries Data Scientist

Brisbane, Australia | +61 493 007 125 | jademaggs@gmail.com | [LinkedIn](#) | [ResearchGate](#) | [Scopus h-index: 13](#) | [GitHub](#) | jademaggs.github.io

I am a fisheries scientist and data professional with 18 years of experience across South Africa, New Zealand, and Australia. I specialise in recreational fisheries survey design and harvest estimation, including the design and delivery of national-scale boat ramp creel survey programmes, fisheries database development, spatial analysis and mapping, and data analysis and modelling. I work primarily in R, SQL, and QGIS. I am registered under an Australian ABN and now available for short-term remote consulting engagements, with a particular interest in Pacific fisheries data and monitoring projects.

Professional Experience

DEPARTMENT OF PRIMARY INDUSTRIES (Brisbane)

Dec 2025 – present

- Developing and maintaining Power Automate flows to deliver data collection summaries via Power BI dashboards
- Building automated data tracking workflows using Power Automate to query databases, update SharePoint lists, run data quality checks via Power BI paginated reports, distribute outlier reports
- Supporting data migration testing with Jira as part of a transition from on-premises MS SQL Server to cloud infrastructure (Azure, Microsoft Fabric)

EARTH SCIENCES NEW ZEALAND (formerly NIWA)

Jan 2022 – Dec 2025

- Group Manager: Modelling and Recreational Fisheries (five staff)
- Project leader for national boat ramp creel surveys of the recreational fishery and other projects
- Fisheries modelling (GLM, CPUE standardisation, harvest estimation, recreational release mortality)
- Member of the Marine Amateur Fisheries and Inshore Fisheries Assessment working groups
- R, PostgreSQL, Linux, QGIS, GitLab
- Led implementation of electronic data acquisition and streamlined workflows
- Center for Independent Experts reviewer

Apr 2020 – Dec 2021

- Group Manager: Fisheries Data Services (six staff)
- Managed all fisheries data from research and observer programmes for NIWA and New Zealand's Ministry for Primary Industries
- Managed commercial rock lobster catch sampling project
- Led development of data loading software and PostgreSQL databases
- Delivered efficiencies and significant improvements across multiple workflows
- PostgreSQL, Bash, Linux, QGIS, GitLab

MINISTRY FOR PRIMARY INDUSTRIES (New Zealand)

Jan 2019 – Mar 2020

- Contributed to the development of ecosystem-based fisheries management approaches
- Member of the Marine Protected Area (MPA) Science Advisory Group
- Engaged with stakeholders on proposed MPA networks, including those affecting commercial rock lobster fisheries

OCEANOGRAPHIC RESEARCH INSTITUTE (South Africa)

May 2006 – Dec 2018

- Led recreational fisheries surveys (shore- and boat-based)
- Fisheries modelling, including GLMs, CPUE standardisation, and fish movement analysis
- Field surveys to monitor relative abundance, movement, and community composition of fishes
- Delivered postgraduate lectures on fisheries stock assessment and R programming
- Performed biological sampling of commercial fish species
- Developed and administered fisheries monitoring databases using PostgreSQL
- Member of the Linefish Scientific Working Group
- Managed a team of three staff and a research aquarium laboratory

Academic Qualifications

- PhD (Fisheries Science) | Rhodes University, 2017
 - Thesis title: Movement of coastal fishery species in southern Africa: research trends, characterisation of behaviours and a case study on fishery implications. Supervisor: Prof. Paul Cowley.
 - MSc (Biology) | University of KwaZulu-Natal, 2011
 - Thesis title: Fish surveys in exploited and protected areas of the Pondoland Marine Protected Area with consideration of the impact of the MPA on coastal fisheries. Supervisors: Dr Bruce Mann & Prof. Paul Cowley.
 - BSc (Zoology, specialising in environmental management) | University of South Africa, 2009
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Lecturing Experience

- Fishery stock assessment (University of KwaZulu-Natal, 2012-2018) - Postgraduate
 - Introduction to R Programming (2016, 2017) - Postgraduate
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Technical Skills

- **Software:** R | Power BI | Power Automate | GIT | QGIS | Linux | Bash | $\LaTeX$
 - **Statistics:** Maximum likelihood | Generalised linear modelling
 - **Data management:** PostgreSQL | MS SQL Server | Azure | Microsoft Fabric | SharePoint | Jira
 - **Field:** Class IV Commercial Diver (563 hrs) | Class IV Commercial Dive Supervisor (313 hrs) | Commercial Boat Skipper | Level 2 First Aid | STCW – PST, PSSR
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Training

- Zonation decision support software training (NIWA, Sep 2019)
 - Ecofish stock assessment interpretation (Dept. of Agriculture, Forestry and Fisheries, Apr 2017)
 - Introduction to $\LaTeX$ (University of KwaZulu-Natal, Jun 2016)
 - Introduction to Fisheries Genetics (University of Stellenbosch, Mar 2016)
 - Species distribution modelling (University of Stellenbosch, Feb 2013)
 - Ecological modelling (IRD/University of Cape Town, Jun 2012)
 - Statistical modelling Using R (Tromsø University, Sep 2011)
 - Fish stock assessment (FAO/SWIOFP - Kenya, Sep-Oct 2010)
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Publications

Career summary: 24 peer-reviewed journal articles, 21 working group presentations, 1 book section, 2 theses, 2 conference proceedings, 81 reports, 22 conference presentations, 2 posters, 16 popular articles. See [ResearchGate](#).
